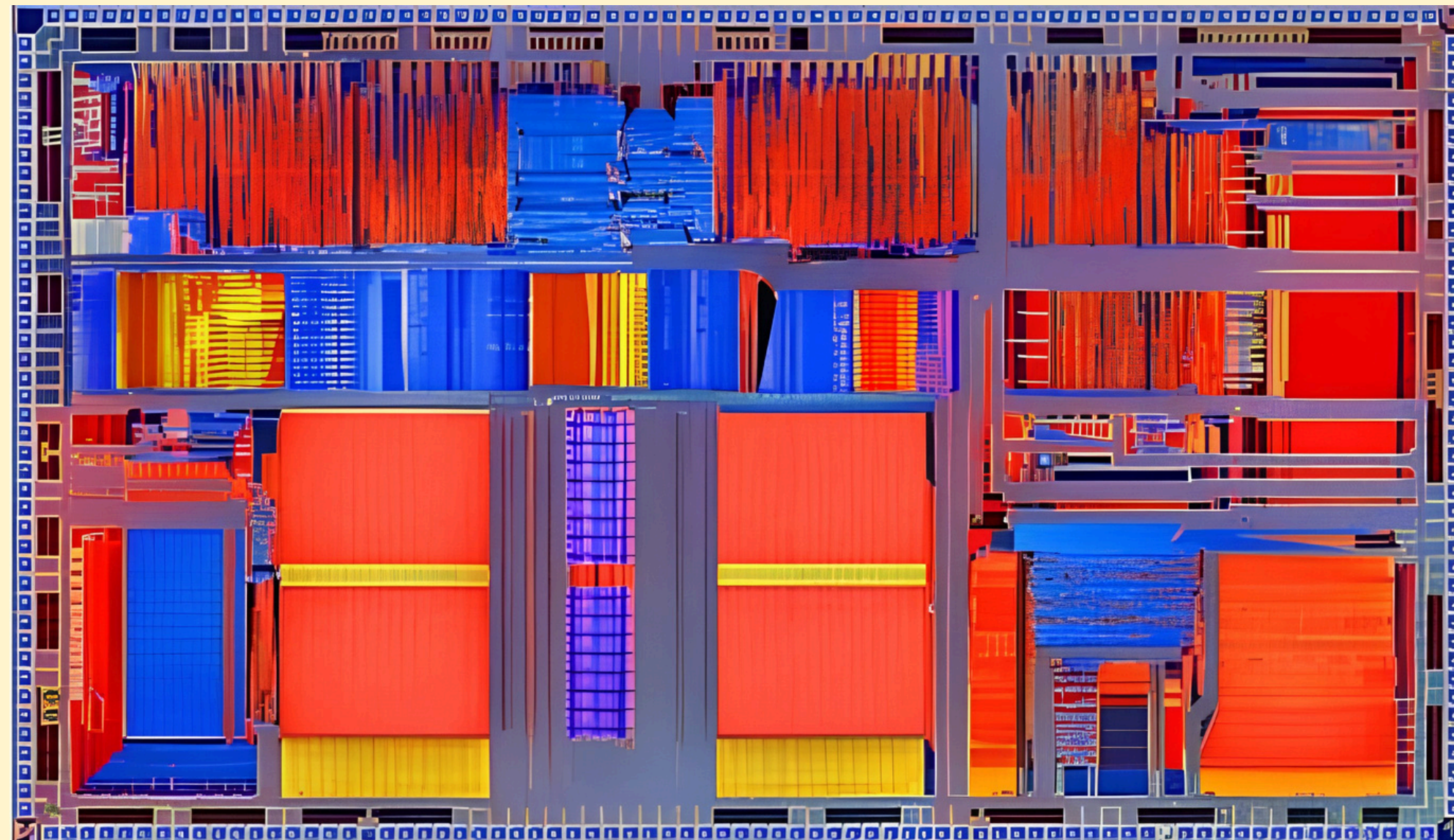


# FLOORPLANNING ARCHITECTURE & IMPLEMENTATION STRATEGIES



PRESENTED BY

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SECTION 01

# Core Sizing & Utilization Targets

Establishing optimal die/core geometry, aspect ratios, target utilization thresholds, and pad-limited vs. core-limited constraints in early Physical Design.

# Factors Influencing Utilization

| Design Factor                     | Low Utilization Strategy (<60%)                | High Utilization Strategy (>75%)                 | Engineering Trade-off / Impact                                                                 |
|-----------------------------------|------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------|
| Advanced Nodes (FinFET / GAA)     | Mandatory due to pin access & track blockage   | High risk of DRC violations & pin congestion     | FinFET grid quantization limits valid cell locations; <b>requires lower density targets.</b>   |
| High-Frequency CPU / GPU          | Required for clock buffer & hold-fix insertion | Incompatible; causes clock skew & delay spikes   | Extra whitespace allows local buffer tree growth and skew balancing during CTS.                |
| Multi-Vt & Setup Optimization     | Allows room for cell upsizing (ULVT/LVT)       | Prevents cell upsizing due to cell overlap       | Timing optimization frequently upsizes driver cells; <b>tight utilization blocks upsizing.</b> |
| Interconnect Layers (Metal Stack) | Used when metal layer count is limited (<6)    | Feasible only with rich metal stack (>10 layers) | Higher metal layer counts enable dense standard cell routing above logic without blocking.     |
| Power Grid Mesh Density           | Compatible with dense PG straps (M6-M9)        | PG straps block standard cell pin access         | Heavy power meshes consume routing tracks, forcing lower effective cell utilization targets.   |

# Die vs Core Sizing & Metrics

## Geometric Definitions

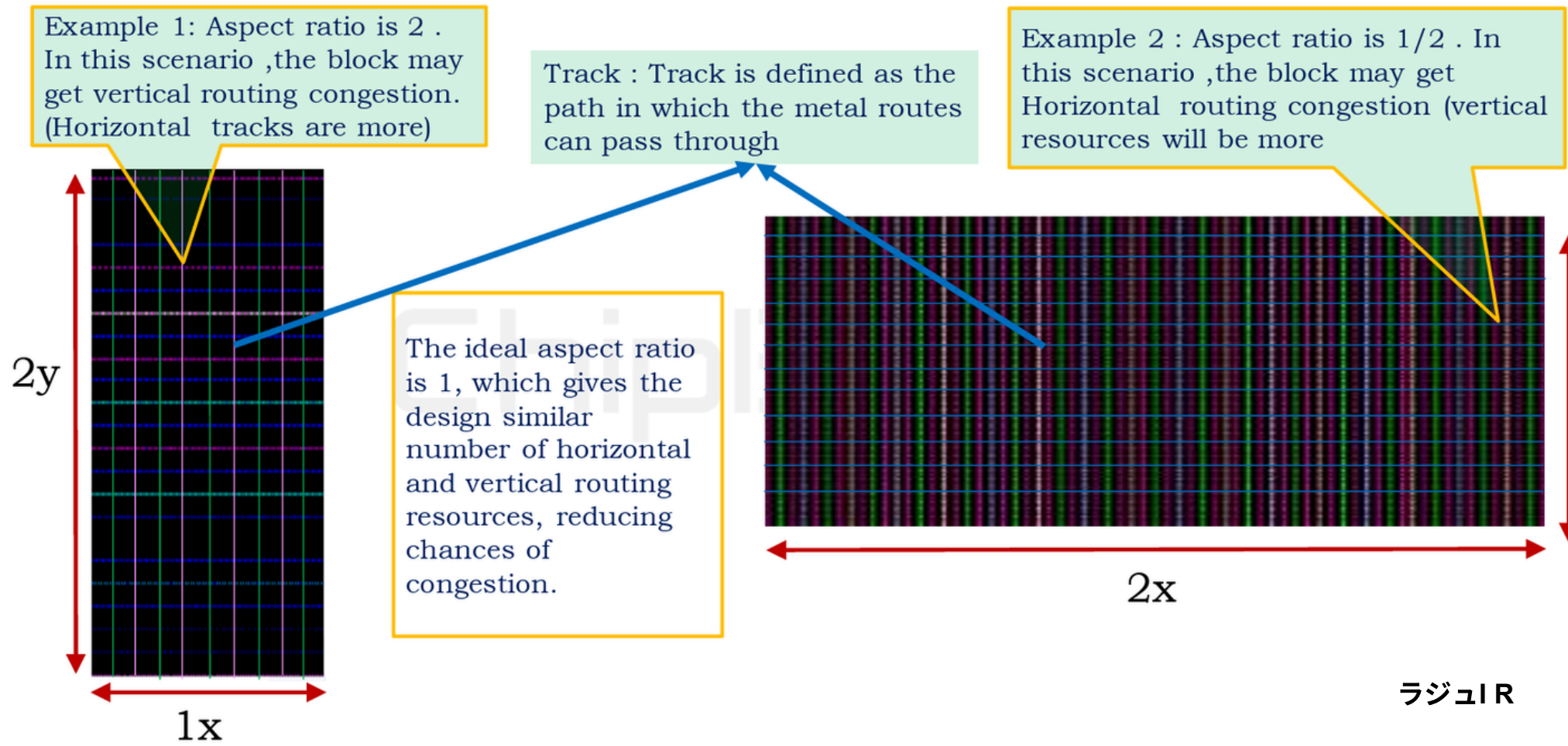
**Die Area:** Total silicon area including the core area, I/O pads, power rings, Electrostatic Discharge (ESD) protection structures, and seal rings.

**Core Area:** Internal bounded area available for standard cell placement, hard macros, IP blocks, and analog blocks.

**Core-to-IO Boundary:** The explicit margin reserved between the outer edge of standard cell rows and the inner edge of I/O pads or die boundary.

**Aspect Ratio Formula:** Defined as Core Height divided by Core Width. Aspect Ratio = 1.0 represents a square core, minimizing average wire length.

**Impact of Aspect Ratio:** Deviations (e.g., 0.5 or 2.0) elongate routing channels in one direction, potentially worsening RC delays across long horizontal or vertical spans



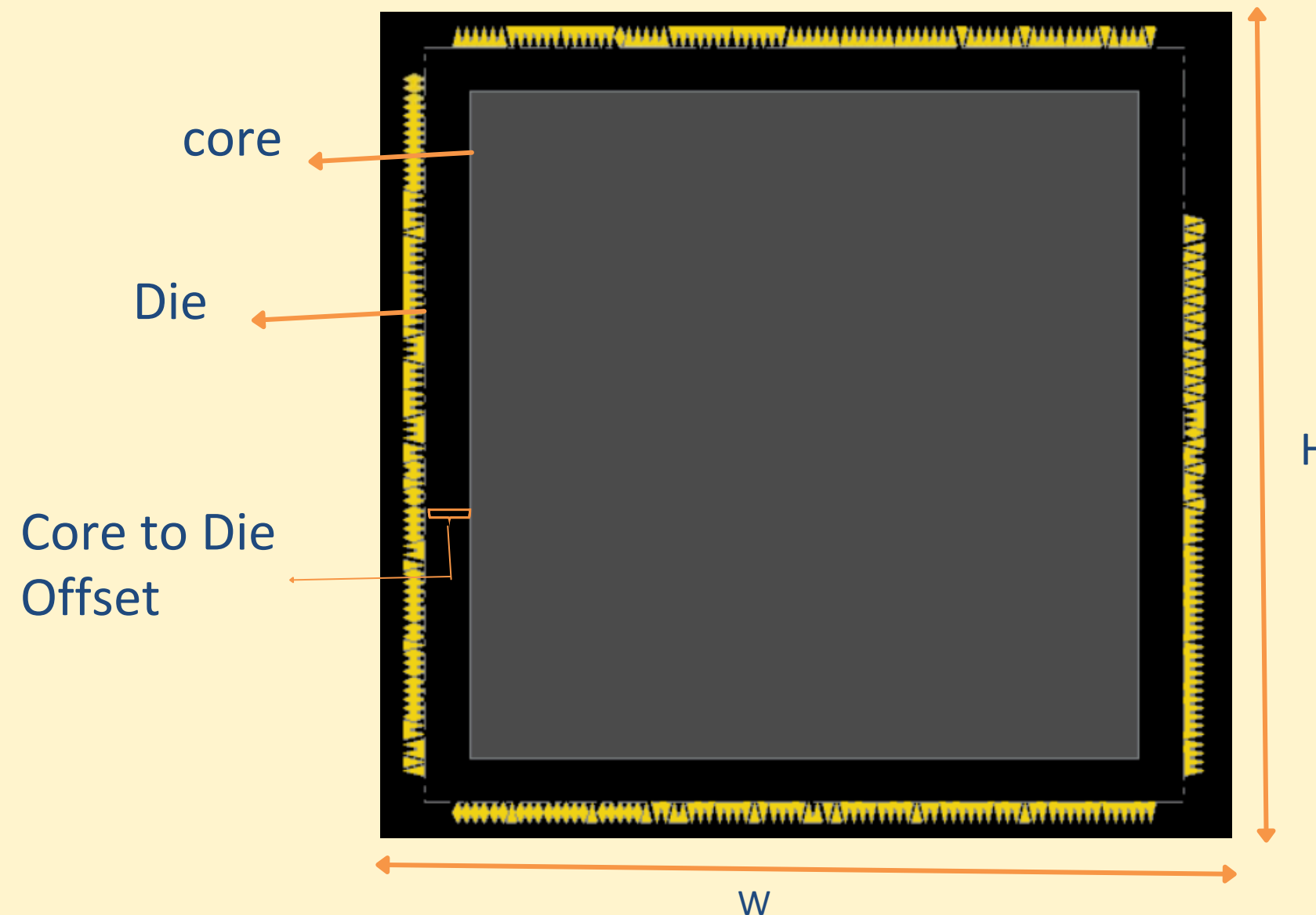
# Limiting Factor Classification

**Core-Limited Designs:** Core area elements (standard cells + macros) dictate total die dimensions. Common in high-density compute/DSP blocks.

**Pad-Limited Designs:** I/O pin count requires a large outer perimeter, leaving excess silicon area inside the core (low core utilization).

**Boundary Margins:** Always specify top, bottom, left, and right core clearance margins (e.g.,  $10\mu\text{m}$  to  $20\mu\text{m}$ ) to accommodate power distribution rings.

**Power Ring Overheads:** Core boundary selection must account for VDD/VSS metal strap widths to avoid cutting into usable placement tracks.



# Utilization Target Analysis

 **Core Utilization Definition:** Expressed as the ratio of active silicon area occupied by standard cells and hard macros to the total usable core area.

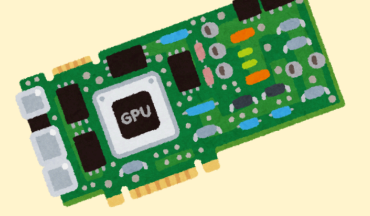
 **Standard Mathematical Formula:**

$$\text{Utilization} = \frac{\text{Area of Standard Cells} + \text{Area of Hard Macros}}{\text{Total Core Area} - \text{Keepout} / \text{Unusable Area}} \times 100\%$$

 **Raw vs. Effective Utilization:** Raw utilization considers total core area, whereas Effective Utilization excludes macro areas, evaluating standard cell density strictly within standard cell placement rows.



# Utilization Target Analysis



- 📈 **Target Ranges by Design Type:** Standard ASIC designs target 65% to 75% initial utilization. High-speed processor cores target 50% to 60% to allow optimization buffers, while dense memories push 80%+.
- ⚠️ **Over-utilization Risks (>80%):** Causes severe placement legalization failures, routing congestion, inability to insert CTS buffers, high cross-talk, and signoff timing closure failure.
- 🛡️ **Under-utilization Risks (<50%):** Leads to silicon area wastage, higher manufacturing cost per die, excessively long wire lengths between distant standard cells, and dynamic power penalties.
- ⚙️ **ECO Reserve Buffer:** Reserve 5% to 8% core area specifically for Engineering Change Orders (ECO) buffer insertion and hold-fix buffers during late-stage timing signoff.

## SECTION 02

# Macro Placement Strategy & Guidelines

Strategic positioning of SRAMs, ROMs, register files, and analog IPs to establish clean routing channels, minimize net wirelength, and prevent congestion hotspots.

# Macro Placement Fundamentals

**Peripheral Macro Placement Rule:** Always place hard macros along the core boundary or corners. Avoid placing macros in the middle of the core area.

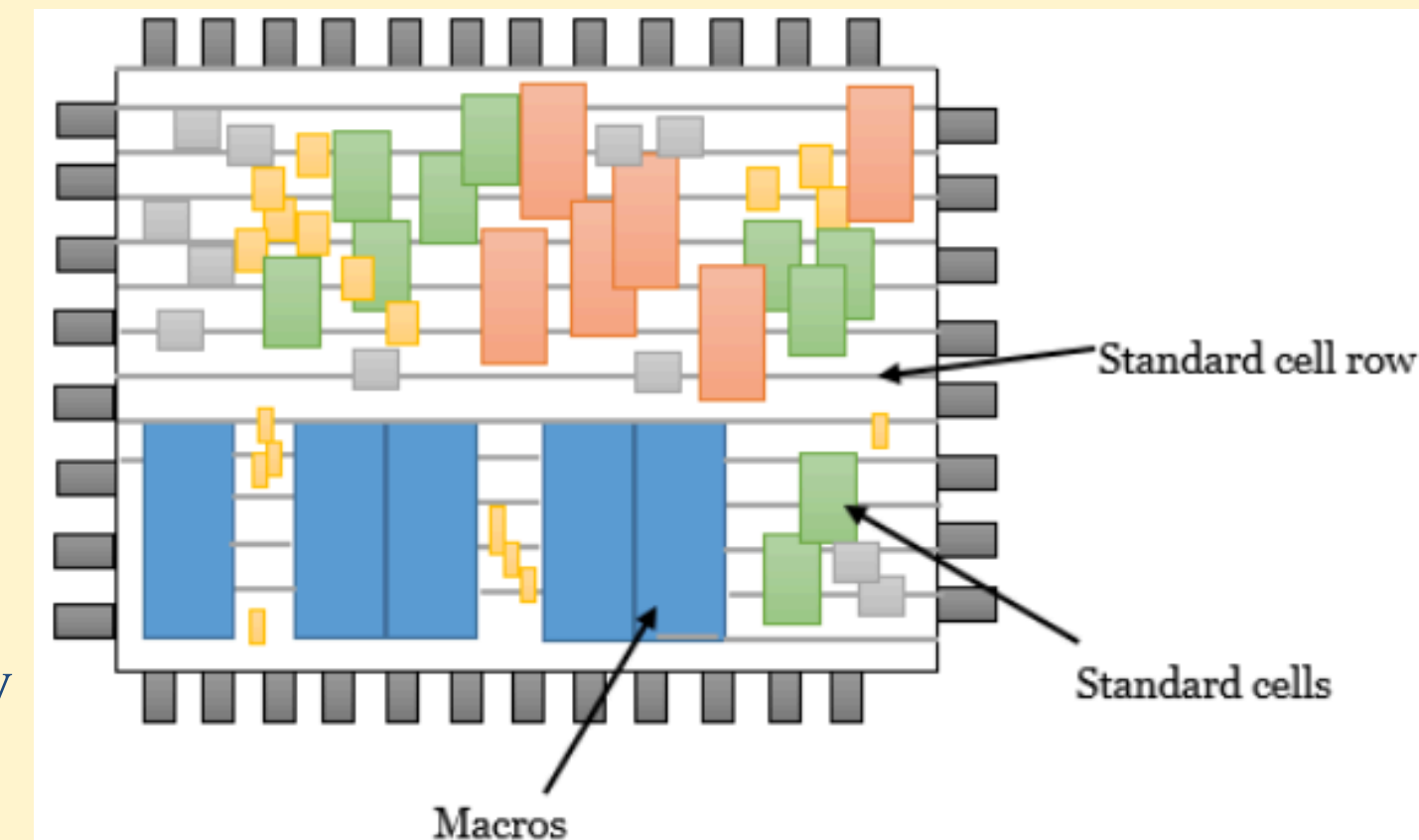
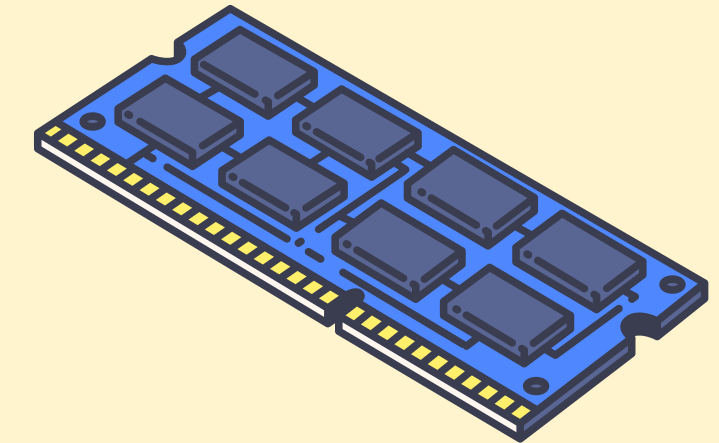
**Core Center Protection:** Leaving the core center free of macros creates a continuous, uninterrupted region for standard cell placement and clock tree distribution.

**Pin Orientation & Facing:** Orient macros such that their signal pins face toward the core center, rather than facing the outer die edge or adjacent macro walls.

**Macro-to-Macro Channels:** Maintain adequate channels between adjacent macros to allow power grid straps and inter-macro routing wires to pass through without bottlenecking.

**Flyline Analysis Driven Placement:** Arrange macros according to logical connectivity to minimize net lengths between controller logic and memory interfaces.

**Criss-Cross Routing Prevention:** Ensure memories that communicate heavily are placed on adjacent edges to prevent long diagonal nets crossing the entire chip.



# Macro Hierarchy & Alignment

## Hierarchical Grouping

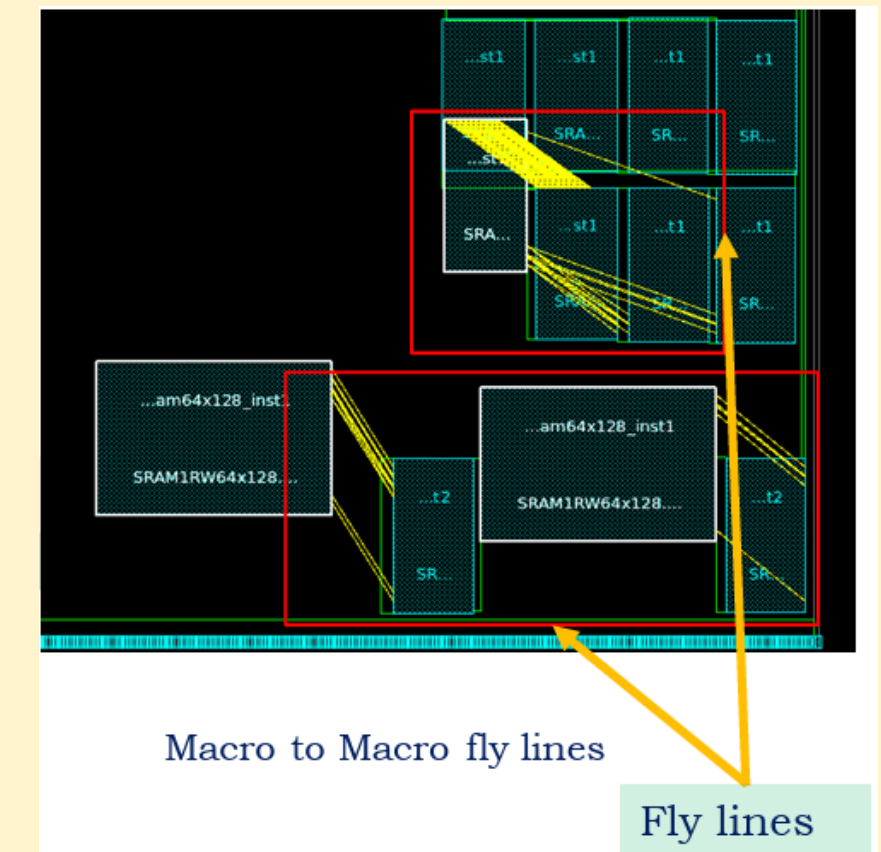
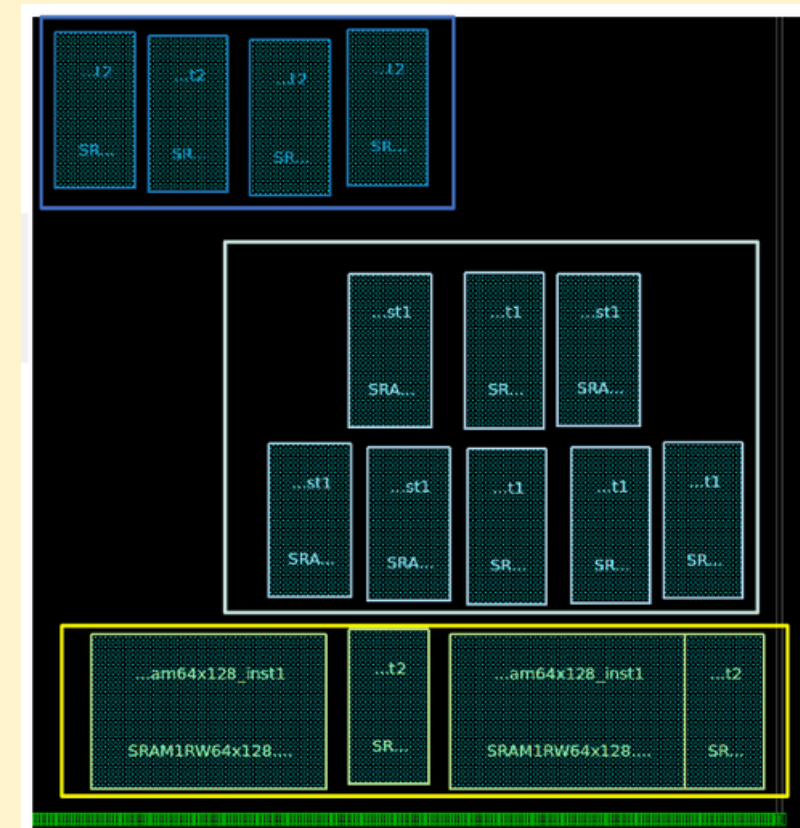
- Group macros belonging to the same logical module (e.g., L1 Cache SRAMs) together in physical space.
- Prevents distributed memory placement which scatters control logic and increases bus latencies.
- Improves floorplan modularity and simplifies hierarchical physical verification and block-level reuse.

## Edge Alignment & Snapping

- Align macros straight along standard cell row grids and manufacturing grid sites.
- Avoid criss-crossed or staggered macro edges that form awkward notch-like whitespace pockets.
- Notches become dead space where standard cells get placed but suffer from severe routing pin isolation.

## Analog & Mixed-Signal IP

- Place noise-sensitive analog IPs (PLLs, ADCs) in dedicated quiet corners away from noisy digital logic.
- Provide dedicated guard rings, independent power pads, and deep N-well isolation around sensitive blocks.
- Prevents digital switching noise from coupling to high-speed clock drivers.



# Macro & Power Grid Co-Design

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## Power Grid Alignment Constraints

- **Stripe Continuity:** Power grid (PG) straps running across the core must connect seamlessly to macro power pins without interruption.
- **Macro Rings:** Large hard macros require dedicated power rings around their perimeter to feed internal memory bitlines and sense amplifiers.
- **Grid Pitch Matching:** Snap macro locations to integer multiples of the power strap pitch to ensure vertical/horizontal PG straps align directly with macro power pads.
- **VSS/VDD Shielding:** High-speed clock pins on macros require adjacent VSS shielding straps integrated during early power planning.

## Thermal & IR Drop Considerations

- **Thermal Hotspot Mitigation:** Clustering multi-port memories in a tight array generates localized thermal hotspots, accelerating electromigration.
- **Thermal Spreading:** Space high-power density memories evenly along die edges to distribute heat dissipation evenly across the silicon substrate.
- **Dynamic IR Drop Control:** High memory read/write switching currents cause dynamic IR drop; place decoupling capacitor (Decap) arrays adjacent to macro corners.
- **Power Shadow Regions:** Regions directly behind large macros may lack PG strap continuity; insert extra power stripes to feed surrounding standard cells.

## SECTION 03

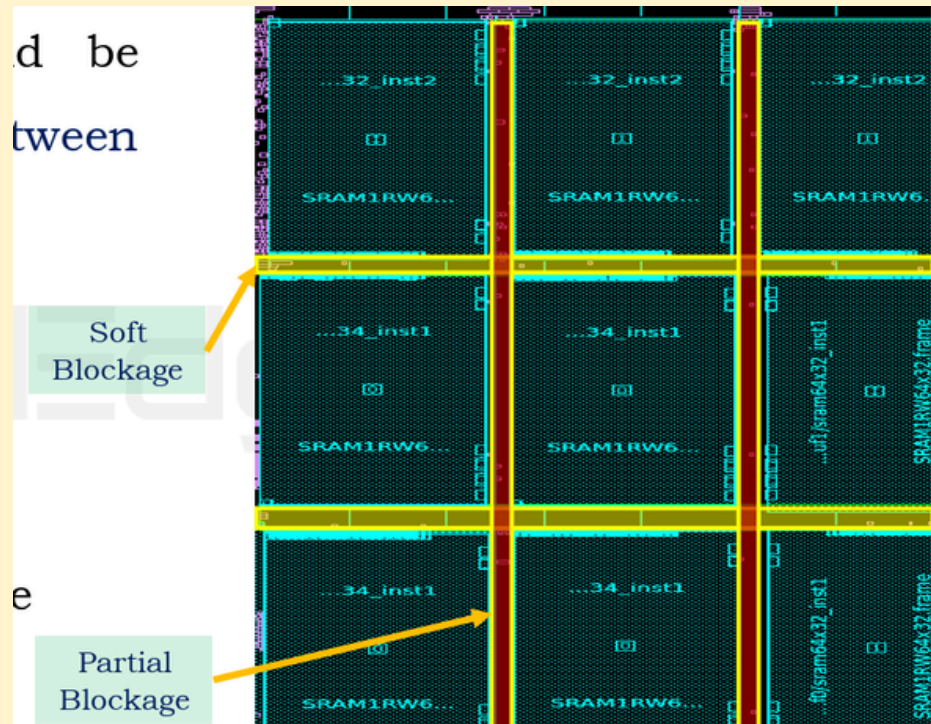
# Halos, Keepout Margins & Blockages

Understanding the distinction between hard, soft, and partial placement blockages, macro keepout halos, and layer-specific routing blockages.

# Placement Blockage Classifications

| Blockage Type              | Standard Cell Placement                   | Buffer / Inverter Insertion                  | Primary Engineering Use Case                                                                        |
|----------------------------|-------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Hard Placement Blockage    | ✗ Strictly Prohibited                     | ✗ Strictly Prohibited                        | Reserving clean channels for power straps, macro interfaces, and physical keepout zones.            |
| Soft Placement Blockage    | ✗ Blocked during Initial Coarse Placement | ✓ Allowed during Optimization & Legalization | Prevents global placer from dumping logic into narrow channels while allowing local timing buffers. |
| Partial Placement Blockage | ✓ Restricted to max % density (e.g., 40%) | ✓ Allowed up to specified density cap        | Spreads out cell density in high-congestion regions near dense macro pin boundaries.                |
| Halo (Macro Keepout)       | ✗ Prohibited around macro perimeter       | ✗ Prohibited (unless halo is soft)           | Moves with the macro dynamically; ensures whitespace around macro edges for DRC/pin access.         |

```
</> Innovus Command syntax: createPlaceBlockage -type hard -box {x1 y1 x2 y2} | addHaloToBlock -allSides 5.0 memory_inst
```



# Halos (Keepout Margins) Strategy

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## Macro Halo Characteristics

- **Definition:** A halo (or keepout margin) is a guard band placed directly around the perimeter of a hard macro block.
- **Dynamic Binding:** Unlike fixed placement blockages, halos are attached to the macro object. Moving the macro automatically relocates the halo.
- **Preventing Pin Congestion:** Standard cells placed too close to macro pins create severe routing grid congestion and cause pin access DRC violations.
- **Power Ring Clearance:** Provides necessary spatial clearance to construct macro power rings without overlapping standard cell power rails.
- **Asymmetric Sizing:** Halos can be specified independently per side (e.g., Left: 3μm, Right: 8μm where pins reside, Top: 2μm, Bottom: 2μm).

## Halo vs Blockage Comparison

- **Static Blockage:** Fixed to core coordinates. If a floorplanner moves a macro, static blockages remain behind, creating erroneous empty spots.
- **Over-constraint Hazard:** Making halos excessively large wastes usable core area, forces standard cells farther away, and increases wire delays.
- **Channel Halos:** When two macros face each other, their individual halos combine to form a clear routing channel (e.g., 5μm + 5μm = 10μm channel).
- **Well-Tap & End-Cap Rules:** Halos prevent standard cell N-well structures from violating macro substrate isolation rules.
- **Recommended Guideline:** Set macro halos equal to at least 2 to 4 times the width/height of standard cell pitch parameters.

# Routing Blockages & Metal Rules

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- 🏗️ **Routing Blockage Definition:** Regions explicitly designated to prevent global and detail routers from placing signal wires on specified metal layers.
- 🛡️ **Macro Internal Layer Blockages:** Hard macros internally utilize lower metal layers (e.g., M1 through M4) for bitlines and wordlines. Routing blockages must cover these layers over the macro footprint.
- ✂️ **Top-Layer Over-Macro Routing:** Upper metal layers (e.g., M7-M9) can often route freely over macros if the macro library DEF provides correct obstruction (OBS) definitions.
- 🔊 **Crosstalk & Noise Prevention:** Apply routing blockages on critical layers directly above sensitive SRAM sense amplifiers to prevent dynamic signal noise injection.
- 🔧 **Via Blockages:** Restrict via stack insertion in tight macro channels to avoid cutting off adjacent metal routing tracks.
- ⚠️ **Antenna Effect Mitigation:** Long routing tracks passing over macros without intermediate via drops accumulate charge, causing gate antenna DRC violations.

## SECTION 04

# Channel-Based vs. Abutted Floorplans

Comparative architectural analysis of channel-based macro arrangements versus modern abutted hierarchical SoC floorplanning strategies.

# Channel-Based Floorplanning

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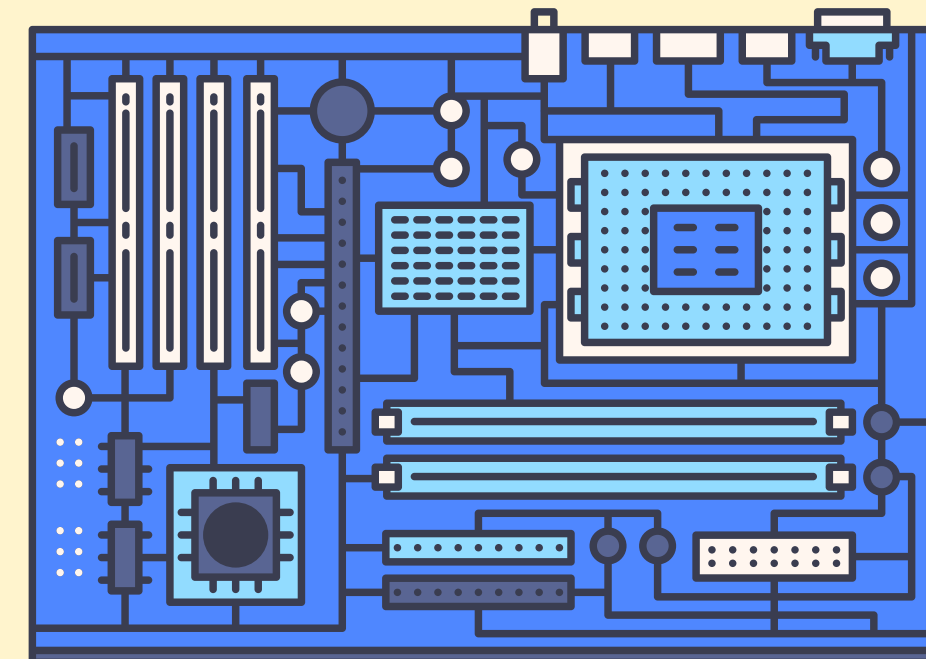
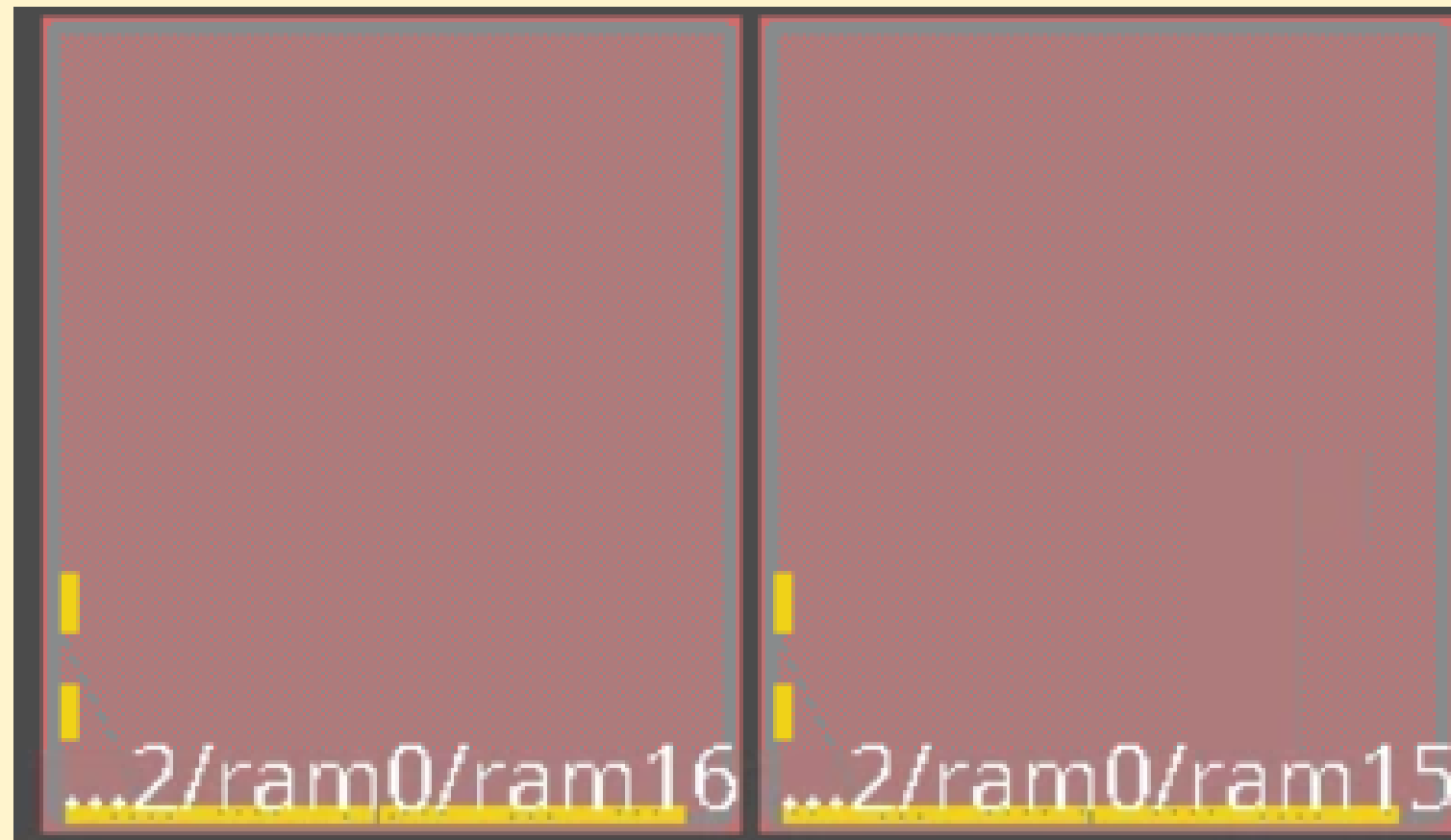
## Architectural Overview

**Definition:** Macros are separated by dedicated open spacing channels containing standard cell rows and routing tracks.

**Routing Access:** Inter-macro signals and global bus lines run directly through the intermediate channel space between macro blocks.

**Buffer Placement:** Channels provide physical space for inserting pipeline buffers, repeaters, and timing closure cells for long inter-block paths.

**Power Distribution:** Allows vertical and horizontal power grid straps to drop down to standard cell rows inside the channels.



# Channel-Based Floorplanning

## Advantages & Drawbacks

### Pros

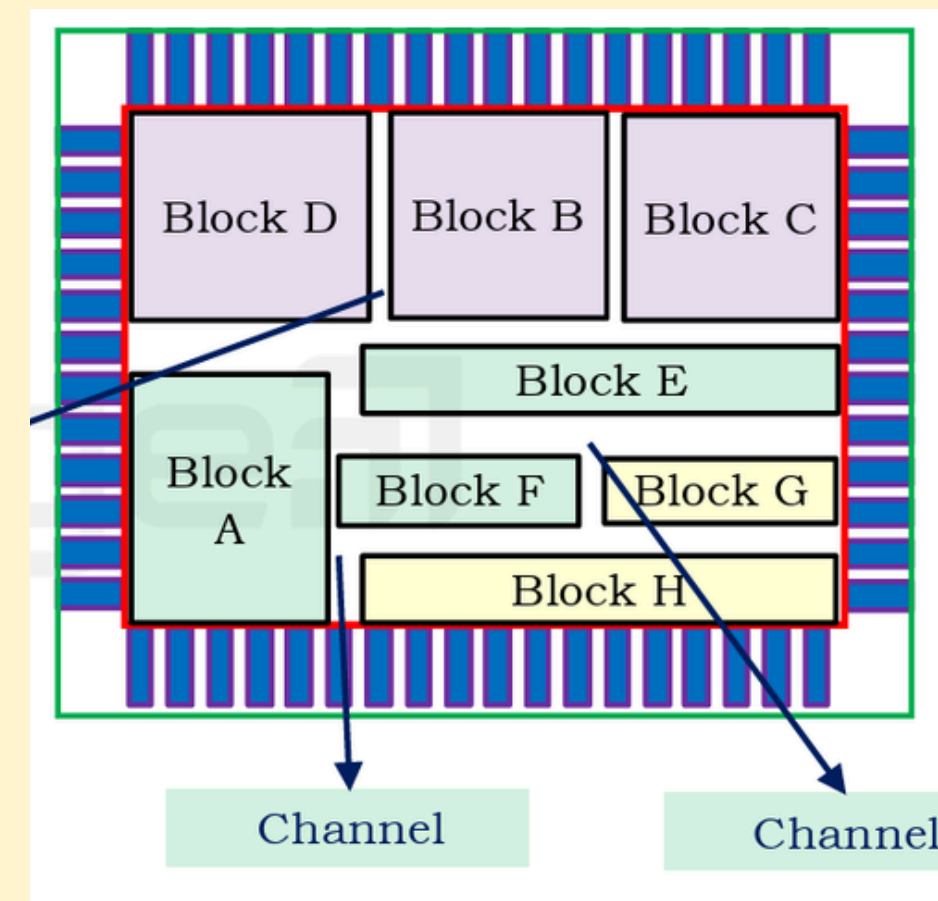
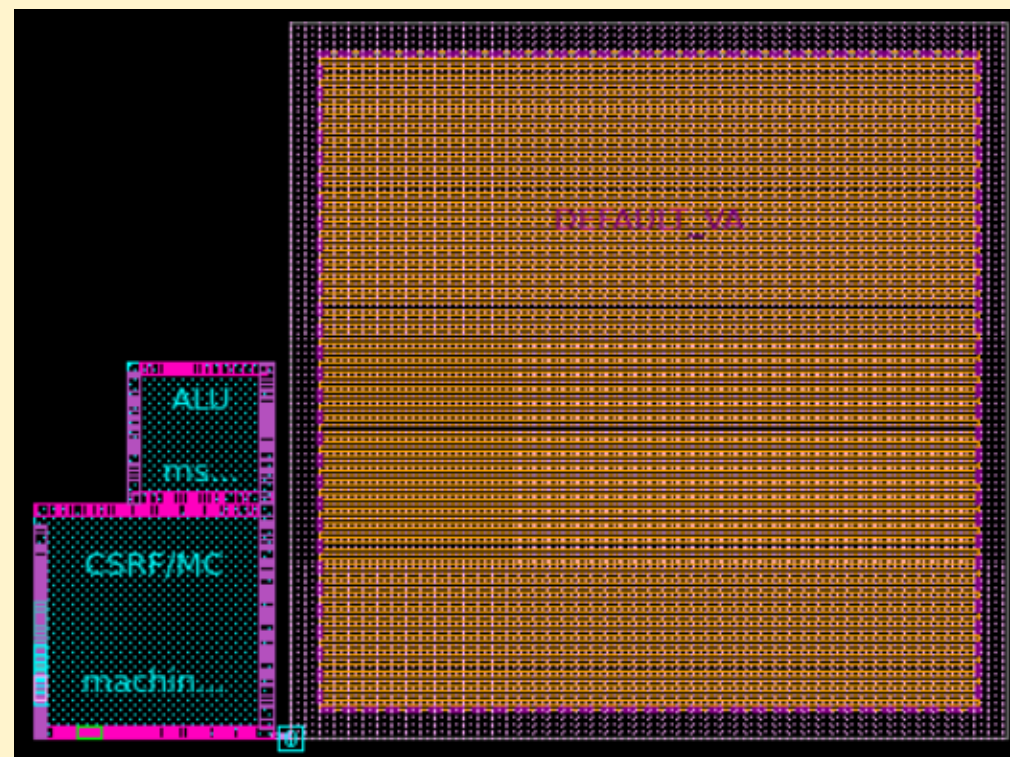
**Lower Congestion:** Spreading macros apart provides abundant routing tracks, preventing severe localized wire bottlenecks.

**Easier Timing Closure:** Allows buffer trees to be placed close to memory interfaces to resolve setup/hold violations.

### Cons

**Area Penalty:** Channels increase total core silicon area, resulting in lower overall die utilization and higher manufacturing costs.

**Long Wire Lengths:** Wire lengths between standard logic and macro pins increase due to physical separation distance.



# Abutted Floorplanning

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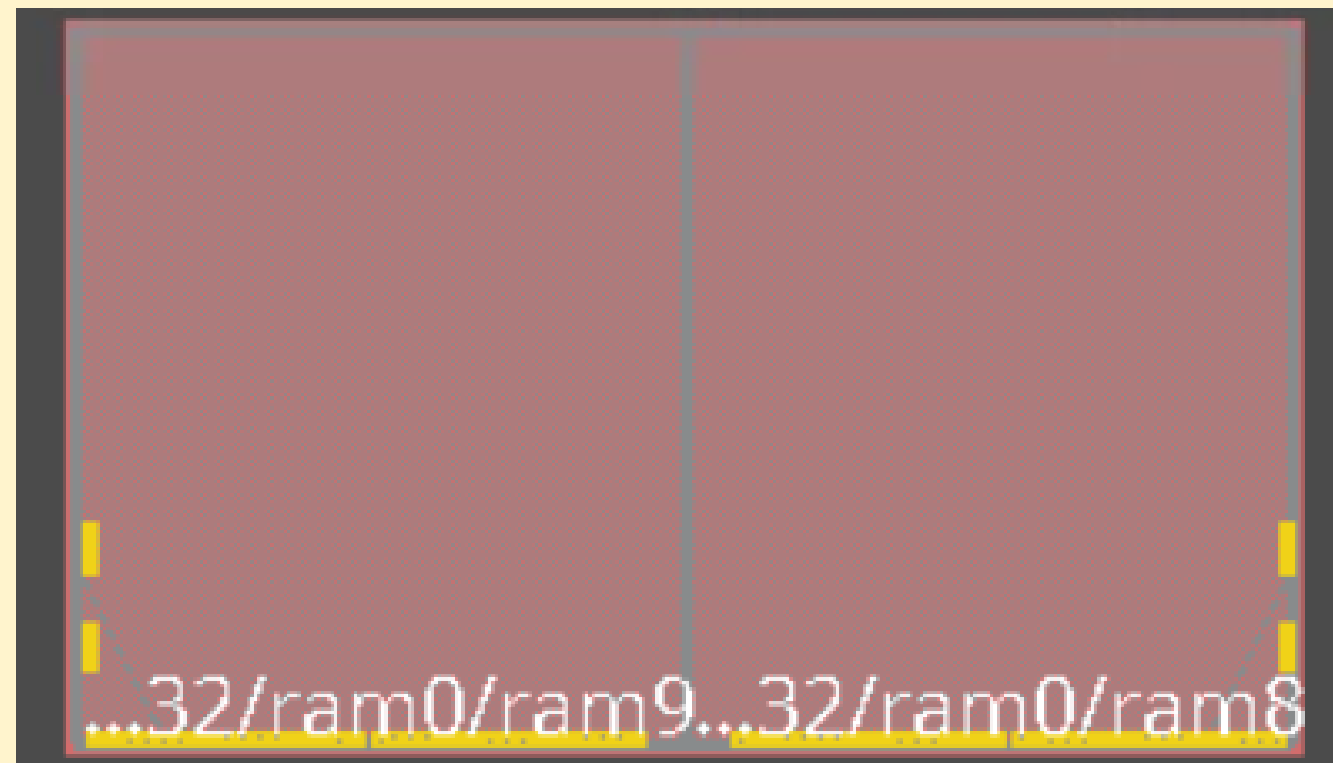
## Abutted Topologies

**Definition:** Macros or hierarchical sub-blocks are placed directly side-by-side with zero physical clearance or channel spacing between them.

**Shared Boundaries:** Abutted blocks share common physical edges, allowing direct pin-to-pin signal connections across boundaries.

**Hierarchical SoC Integration:** Widely used in modern multi-core SoCs where entire functional sub-blocks (cores) abut each other seamlessly.

**Maximum Area Efficiency:** Eliminates wasted whitespace channels, maximizing silicon core utilization and reducing total die footprint.



# Abutted Floorplanning

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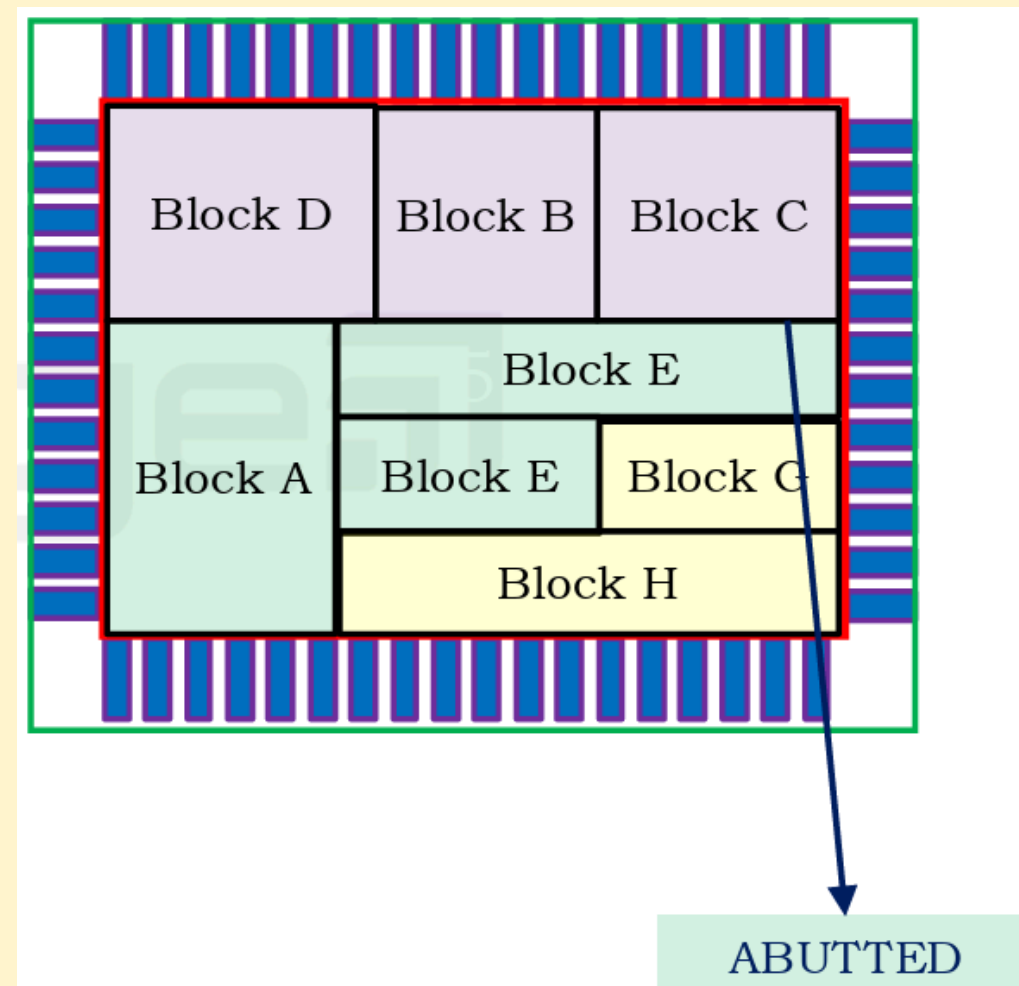
## Implementation Challenges

**Routing Over the Top:** All inter-block signals must route on upper metal layers directly over the abutted macro footprints.

**Pin Alignment:** Pins on abutted block boundaries must match exact coordinates and metal layers to prevent interface DRCs.

**No In-Channel Buffering:** Long nets crossing abutted blocks cannot insert buffers mid-route unless feedthrough tracks are provisioned.

**Power Ring Sharing:** Requires carefully co-designed merged power rings along the common abutment boundary edge.



# Hybrid Floorplanning Trade-offs

| Floorplan Attribute       | Channel-Based Approach         | Abutted Architecture                 | Hybrid Strategy (Industry Standard)                       |
|---------------------------|--------------------------------|--------------------------------------|-----------------------------------------------------------|
| Silicon Area Efficiency   | Low (High whitespace overhead) | Maximum (Minimal whitespace)         | Balanced (Abutted macros in groups + channels for busses) |
| Routing Congestion Risk   | Low to Moderate                | High (Hotspots over macro corners)   | Controlled via targeted partial blockages in channels     |
| Timing Buffer Insertion   | Flexible inside channels       | Restricted                           | Buffer corridors embedded between macro clusters          |
| Power Grid Uniformity     | Continuous across channels     | Disrupted by macro footprints        | Upper layer power mesh spans abutted blocks seamlessly    |
| Implementation Complexity | Simpler physical flow          | Complex pin alignment & feedthroughs | Optimal balance for sub-5nm complex SoCs                  |

Best Practice: Abut macros belonging to the same memory cluster (e.g. dual-port SRAM pairs), but leave 15-20µm corridors between distinct functional clusters.

SECTION 05

# Pin / I/O Planning & Net Connectivity

Optimal pin distribution, metal layer assignment, edge spacing, and flyline connectivity  
visualization to minimize global wire routing criss-cross.

# Pin and I/O Planning Mechanics

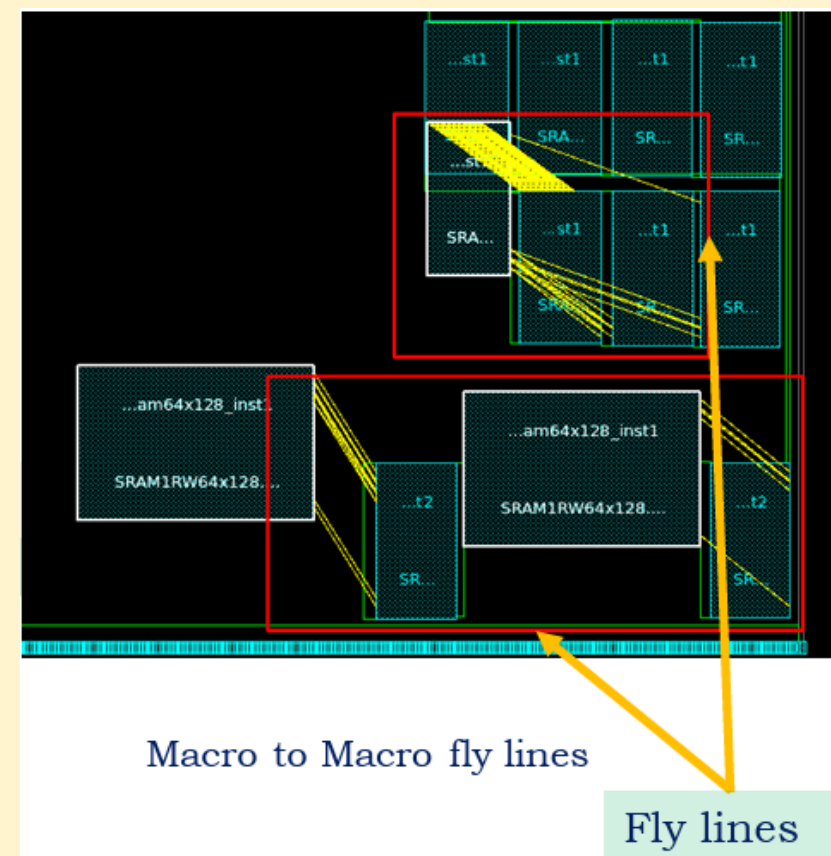
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- 📍 **I/O Pin Placement Objectives:** Map logical top-level signals to physical core boundary locations to minimize total interconnect wirelength and delay.
- 🏗️ **Metal Layer Assignment:** Place high-speed critical I/O pins on higher, low-resistance metal layers (e.g., M5/M6), reserving lower layers for local standard cell pins.
- ↔️ **Pin Spacing & Density Control:** Enforce pitch constraints to avoid clustering hundreds of pins in one narrow boundary segment, which creates severe local routing bottlenecks.
- 🕒 **Clock & Reset Pin Placement:** Position global clock inputs and reset pins centrally along the core boundary to ensure balanced driver distribution to internal clock trees.
- 🛡️ **Bus Pin Grouping:** Keep multi-bit bus pins (e.g., 64-bit data bus) sequential and grouped together on the same metal layer to prevent signal skew and bit-line routing crossovers.
- ⚡ **ESD and Guard Ring Separation:** Maintain mandatory physical distance between sensitive high-speed analog I/O pins and noisy digital supply pads.

# Flyline & Netlist Analysis

## Understanding Flyline Analysis

- **Definition:** Flylines are straight visual connection vectors drawn between block/macro pins based on netlist connectivity before physical routing takes place.
- **Density Visualization:** Heavy bundles of intersecting flylines highlight high interconnect traffic zones and potential routing congestion hotspots.
- **Iterative Macro Floorplanning:** Shift and flip macros based on flyline direction; align macro pins directly opposite the standard cell logic blocks that drive them.
- **Center-of-Gravity Placement:** Place macros near the "center of gravity" of their connected logical modules to minimize cumulative wire length.

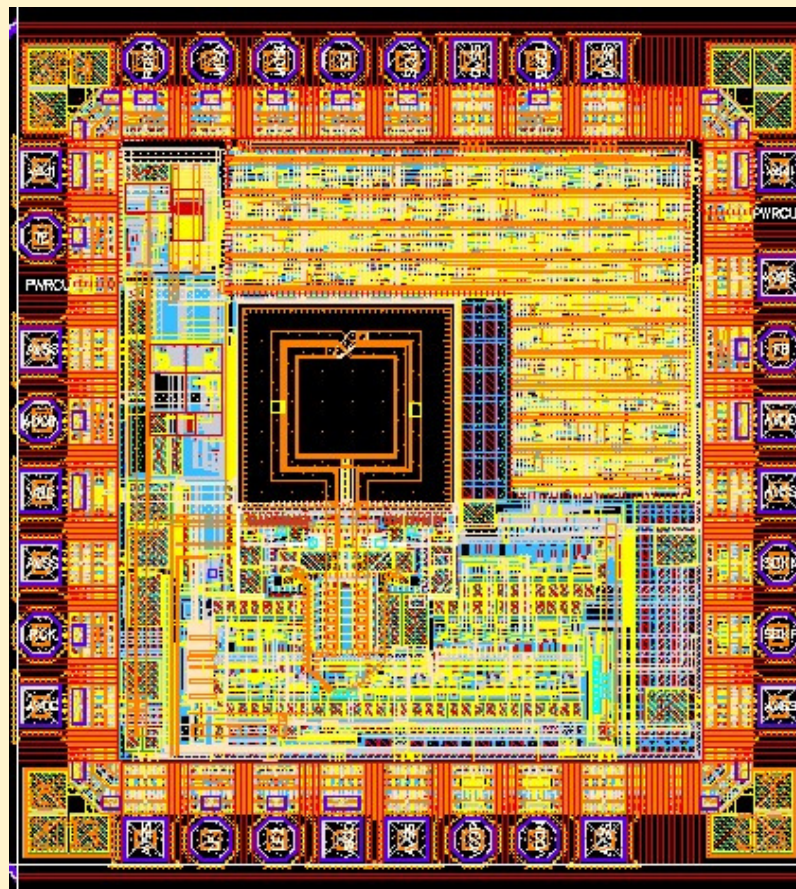


# Flyline & Netlist Analysis

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## Macro Orientation Optimization

- **Rotation Transformations:** EDA tools support 8 orientation states (R0, R90, R180, R270, MX, MY, MX90, MY90) to flip macro pin sides.
- **Untangling Flylines:** Flipping can move its pin interface from an outside boundary wall to directly face the internal core logic.
- **Eliminating Wire Loops:** Proper orientation prevents signal tracks from looping around the backside of a macro footprint to reach the pin interface.
- **Quantifiable Metric:** Measure total Weighted Wire Length (WWL) across all flylines to benchmark floorplan quality prior to running placement.



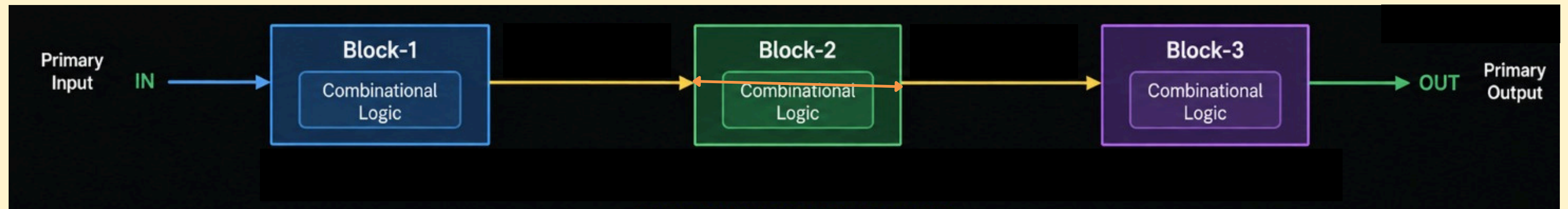
# Feedthrough Mechanics & Buffering

## Feedthrough Definition & Need

- **Concept:** A feedthrough occurs when a signal wire originating in Block A must physically pass through Block B to reach its final destination in Block C.
- **Hierarchical Necessity:** Essential in large abutted SoCs where routing around the perimeter of large intermediate blocks adds unacceptable wire delay.
- **Over-the-Block Routing:** Wires route across upper metal layers above the intermediate block without logically connecting to that block's internal circuits.
- **Ports & Pin Creation:** Creates virtual feedthrough I/O ports on the boundaries of intermediate blocks during top-level floorplanning.

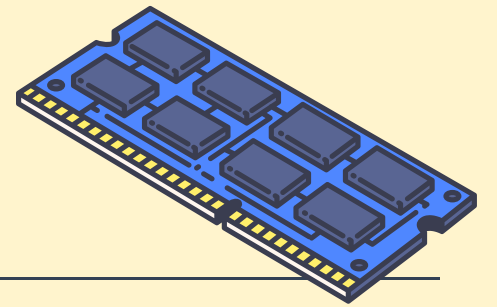
## Buffer Insertion & RC Degradation

- **Long Wire RC Penalty:** Unbuffered feedthrough wires spanning several millimeters suffer severe RC delay degradation and transition edge slurring.
- **Feedthrough Buffer Insert:** Insert repeaters inside the intermediate block specifically reserved for driving top-level feedthrough signals.
- **Keepout Protection:** Protect feedthrough buffers from being optimized away or moved by block-level EDA synthesis and placement tools.



# Dedicated Channels vs Tracks

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- = **Dedicated Feedthrough Corridors:** Creating physical narrow corridors (standard cell placement rows) sandwiched between hard macros dedicated purely to feedthrough signals and repeaters.
- ≡ **Feedthrough Track Allocation:** Reserving specific metal routing tracks on M5, M6, or M7 layers across sub-blocks exclusively for top-level global nets.
- 🔒 **Blockage Protection:** Applying feedthrough routing blockages to prevent block-level local net routers from consuming reserved global feedthrough tracks.
- 🕒 **Timing Budget Assignment:** Top-level implementation must allocate explicit setup/hold timing budgets to the feedthrough segment inside intermediate blocks.
- ⚠️ **Crosstalk Isolation:** Shield feedthrough bus tracks with parallel ground lines (VSS tracing) to prevent switching noise coupling into internal block logic.
- 🔗 **Top-Block Co-Design:** Requires tight synchronization between top-level floorplanner and block-level owners to maintain pin alignment stability.

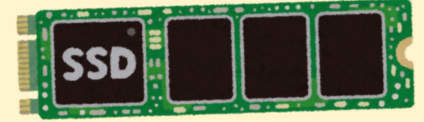
SECTION 06

**Cost of a Poor Floorplan**

**Downstream Impact on CTS & Routing**

Quantifying the severe engineering overhead, timing failure, IR drop penalty, and routing congestion caused by an sub-optimal floorplan during CTS and detailed routing.

# Floorplan Impact on CTS Quality



## Clock Skew & Insertion Delay

- **Unbalanced Tree Sinks:** Asymmetric or scattered macro placements force clock trees to stretch across long distances, leading to massive clock insertion delays.
- **High Clock Skew:** Severe physical displacement of clocked memory pins makes skew balancing between standard cells and macros near impossible.
- **Clock Tree Buffer Explosion:** CTS tools insert excessive clock buffers to drive long interconnects, driving up dynamic clock power dissipation.
- **Clock Routing Congestion:** Clustered macro clock pins choke available upper metal routing channels needed for wide clock trunk lines.

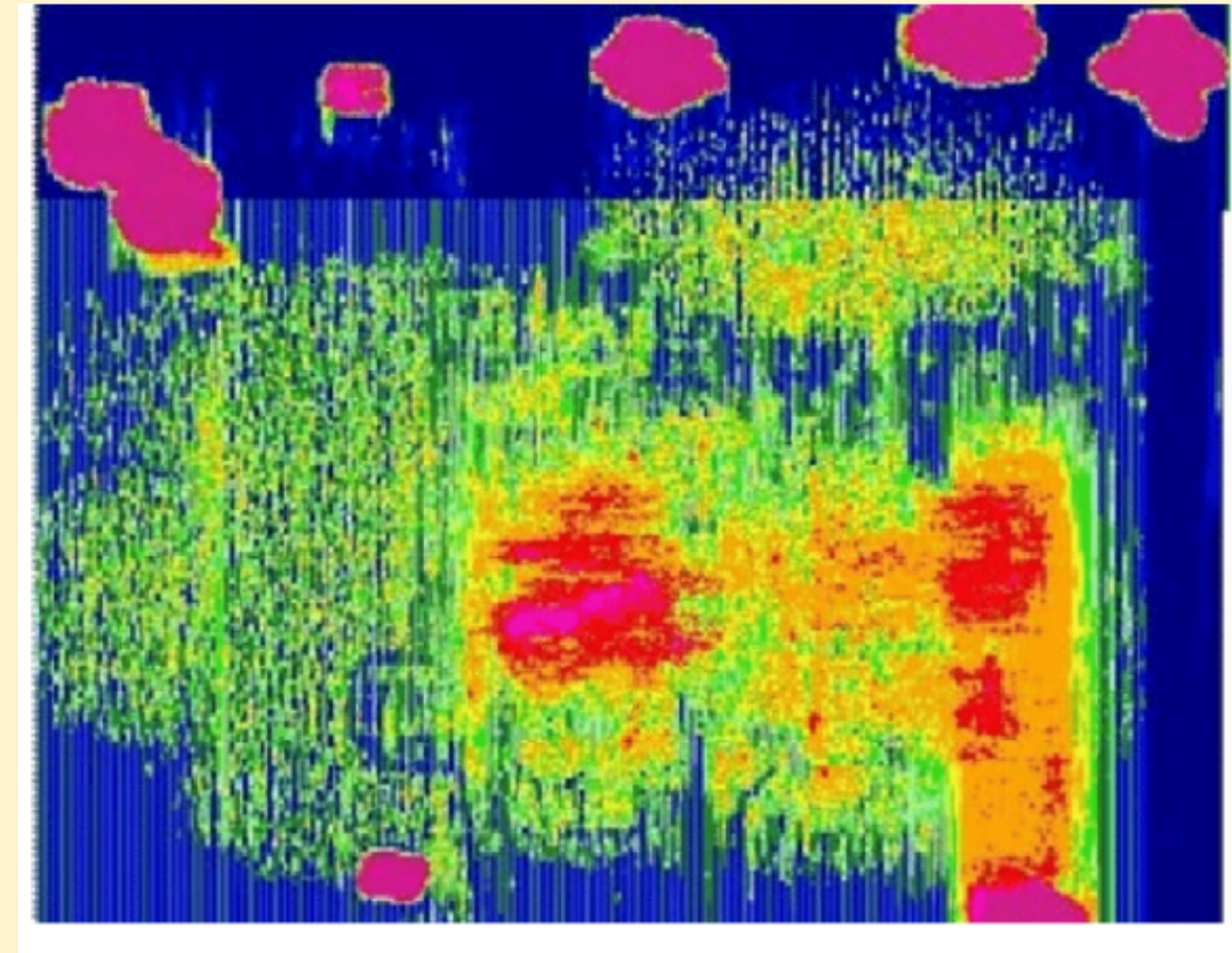
## Power & Timing Degradation

- **Dynamic Clock Power Penalty:** Clock network accounts for 30-40% of total SoC power. A bad floorplan can increase clock power by 25%+ due to extra buffers.
- **Hold Time Violation:** Huge clock insertion delays worsen hold time violations across corner modes, requiring thousands of hold-fixing buffers.
- **Duty Cycle Distortion:** Passing clock trunks through narrow macro channels increases duty cycle degradation.
- **CTS Signoff Failure:** Unable to achieve target skew (<50ps) and max latency, forcing a complete floorplan iteration loop.

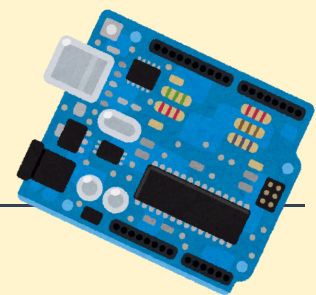
# Routing Congestion & Detours

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- 🔥 **Routing Hotspots & Overflow:** Poor macro placement creates narrow channel bottlenecks where routing demand exceeds available metal tracks (overflow > 0).
- 📡 **Excessive Detour Routing:** Router is forced to route signal wires around huge macro blockages, inflating wire lengths by 200% to 300%.
- ⚠️ **DRC Explosion:** Unresolvable short circuits, minimum spacing, and pin access DRC violations near dense macro pin boundaries.
- 🕒 **Setup Timing Collapse:** Detour wire lengths increase parasitic capacitance (C) and resistance (R), destroying setup timing critical paths.
- ⌛ **Run-time Inflation:** Detail router spends days looping endlessly in attempt to resolve local track congestion, stalling tapeout schedules.



# Floorplan Quality Signoff Matrix



| Floorplan Check Item       | Target Threshold / Rule                 | Verification Command / Tool       | Pass/Fail Signoff Criteria             |
|----------------------------|-----------------------------------------|-----------------------------------|----------------------------------------|
| Effective Core Utilization | 60% - 70% (Node dependent)              | check_design / report_utilization | No local density hotspots > 85%        |
| Macro Halos & Keepouts     | 5µm - 15µm on all pin-facing sides      | check_floorplan -halos            | Zero standard cell overlap with halos  |
| Global Routing Congestion  | Max H/V Overflow < 0.5%                 | check_congestion / trial_route    | No severe GCELL congestion clusters    |
| Macro Pin Alignment        | Pins facing core center & track snapped | check_pin_assignment              | Zero pin access or grid blockage DRCs  |
| Power Grid Continuity      | Static IR Drop < 2% of VDD              | Voltus / RedHawk IR Analysis      | No power strap cuts or ungrounded pins |
| Flyline Crossings          | Minimal weighted wire length            | display_flylines -density         | Parallel bus alignments validated      |

**THANK YOU**